

Multi-Faceted Interpolation

Classic OT

Sinkhorn Divergence

MMD

Computational
Complexity

$$\mathcal{O}(n^3 \log n)$$

$$\mathcal{O}(n^2/\varepsilon^2)$$

$$n^2$$

Statistical
Complexity

$$\mathcal{O}(n^{-\frac{1}{d}})$$

$$\mathcal{O}(\epsilon^{-\frac{d}{2}} n^{-\frac{1}{2}})$$

$$\mathcal{O}(n^{-\frac{1}{2}})$$

Coupling

Hard/sparse
coupling

Soft/dense
coupling

Naive (independent)
coupling

Geometry

Via cost

Via cost

Via kernel

Application: SD for Generative Modeling

Learning Generative Models with Sinkhorn Divergences

Aude Genevay
CEREMADE,
Université Paris-Dauphine

Gabriel Peyré
CNRS and DMA,
École Normale Supérieure

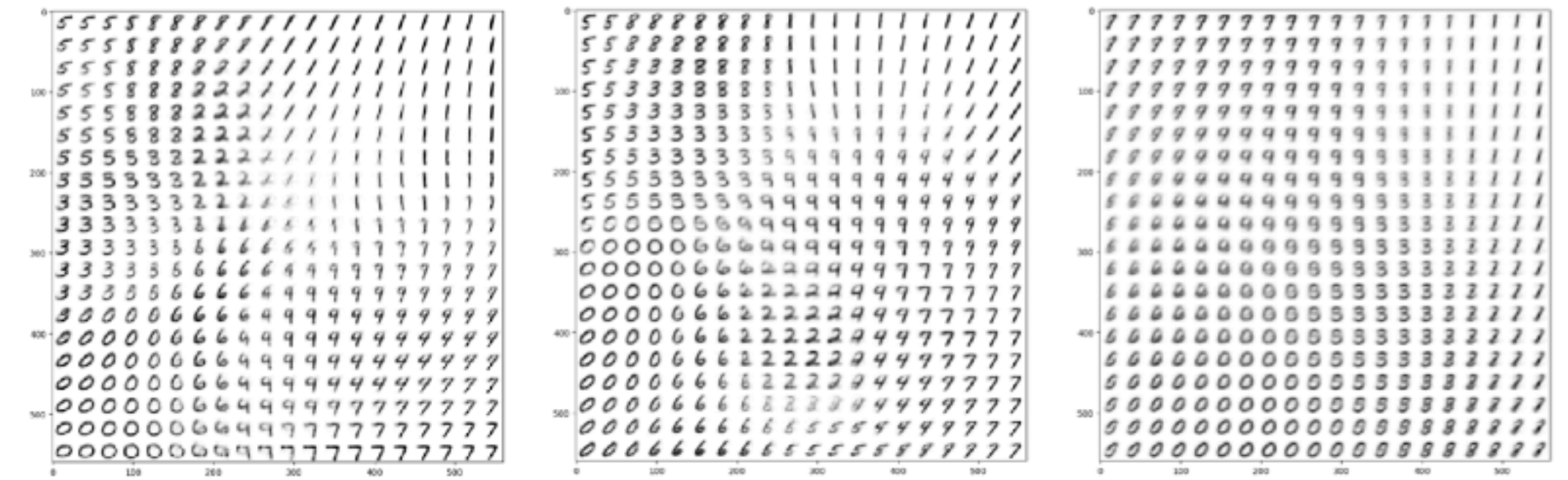
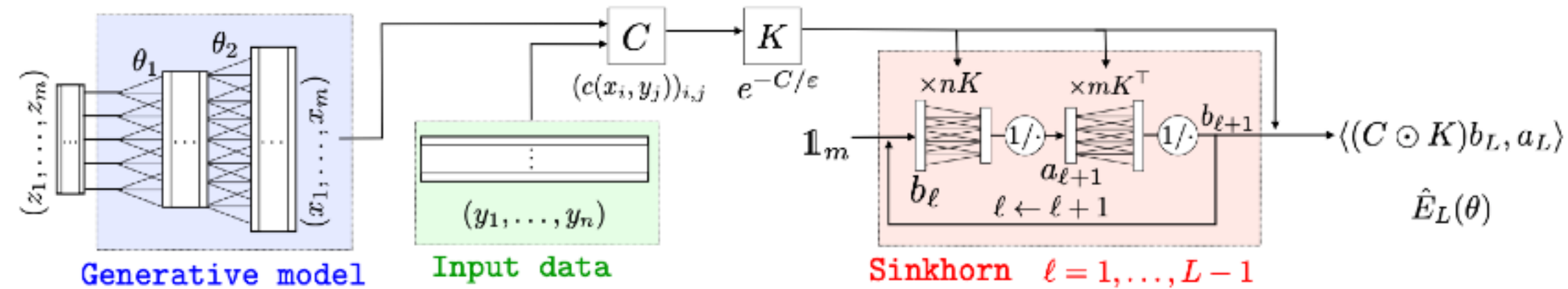
Marco Cuturi
CREST ENSAE
Université Paris-Saclay

Abstract

The ability to compare two degenerate probability distributions, that is two distributions supported on low-dimensional manifolds in much higher-dimensional spaces, is a crucial

1 Introduction

Several important statistical problems boil down to fitting densities, *i.e.* estimating the parameters of a chosen model that *fits* observed data in some meaningful way. While the standard approach is maximum likelihood estimation, this approach is often flawed in

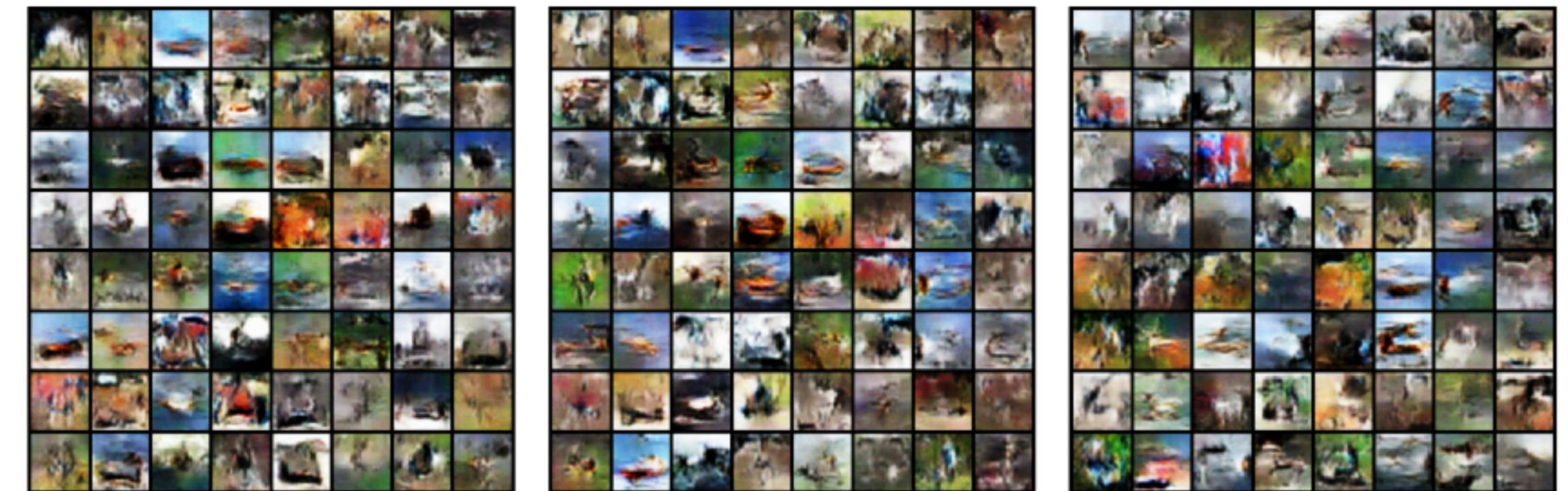


(a) $\epsilon = 1$
 $m = 200, L = 10$

(b) $\epsilon = 10^{-1}$
 $m = 200, L = 100$

(c) $\epsilon = 10^{-1}$
 $m = 10, L = 300$

Figure 3: Influence of the hyperparameters on the manifold of generated digits.



(a) MMD

(b) $\epsilon = 1000$

(c) $\epsilon = 10$